



Disease Surveillance & Early Outbreak Detection

How to Use the Platform

A step-by-step guide for everyday users — no technical background needed

Powered by an Adaptive SVM model for intelligent, self-learning surveillance

Open the platform:

disease-surveillance-early-outbreak-detection.streamlit.app

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Operator's Manual · 2026

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1. What this platform does

This platform is an early-warning system for disease outbreaks in Nigeria. It quietly does two everyday jobs and keeps learning as it goes:

- **Helps assess a patient.** When someone comes in unwell, you enter their symptoms and the system instantly suggests the most likely condition (and other possibilities to consider).
- **Watches for outbreaks.** It tracks confirmed case numbers across all 36 states and the FCT, and highlights the states where a disease looks like it may be flaring up — so officers can act early instead of finding out late.

Because it is built on an adaptive (self-learning) model, every new case you enter teaches it a little more. It does not go stale the way a fixed report would.

Important: This is a decision-support and screening tool. It points you in the right direction quickly; it does not replace laboratory testing or a clinician's judgement. Always confirm before acting on a diagnosis.

2. Getting started

There is nothing to install. Open the link on any phone, tablet, or computer browser:

<https://disease-surveillance-early-outbreak-detection.streamlit.app>

The screen has two parts:

- **The menu — on the left** (the “sidebar”). This is how you move between the different pages.
- **The working area — on the right.** This shows whatever page you have chosen from the menu.

At the bottom of the left menu you will also see a set of small status lights and one switch:

The status lights

Light	What it tells you
 Case records	The system is saving the cases you register. Green means everything is being recorded normally.
Documents & notes	Whether the extra store for clinical notes, lab results and uploaded documents is connected. Green = connected; grey = running without it (cases still save).
Live updates	Whether the live case feed is connected. Green = on; grey = off. The platform works either way.

Tip: A grey light is not an error. It simply means an optional extra is switched off. The core of the platform — registering cases and monitoring outbreaks — always works.

The Proactive alerting switch

This one switch decides how the platform sends warnings. You will learn exactly what it does in Section 6; for now, just know it lives here at the bottom of the menu.

3. The menu at a glance

Six pages, each with one clear job:

Menu item	Use it to...
 Overview	See the whole picture at a glance — how many diseases are watched, how many cases, how far back the data goes.
 Outbreak Monitor	Pick a disease and see which states are most at risk right now, with a recommended action for each.
 Register a Case	Enter a patient's symptoms and details, get an instant assessment, and save the case.
 Trends	See how cases, deaths, risk or alerts have moved over time for a disease and a state.
 Alerts	Read the log of warnings that have been sent to the response officers, and tick them off.
 Model	See exactly what the system covers and how well it performs.

4. Using each page, step by step

4.1 Overview — the big picture

This is your home page. The row of cards across the top gives you the state of the whole system at a glance:

Card	What it means
Diseases monitored	How many diseases the outbreak side is watching. Hover over it to see how many use real data versus demonstration data.
Conditions triaged	How many conditions the symptom-checker can recognise (41).
Total surveillance cases	The total number of recorded cases across every disease.
Data period	The span of dates the records cover (for example, 1971–2026).
States / regions	How many states and regions are represented.
Cases registered here	How many cases have been entered through this platform so far.
Alerts logged	How many warnings have been recorded.
Outbreak accuracy	How well the outbreak model flags real outbreaks (about 89%).

4.2 Outbreak Monitor — where is an outbreak starting?

1. **Pick a disease** — use the “Disease” box at the top. The list includes every disease being watched.
2. **Read the risk table** — each state is listed with its recent cases, a risk reading, a trend arrow, and a recommended action.
3. **Use the colour to prioritise** — the three cards above the table count how many states are HIGH, MEDIUM and LOW risk.

What the colours mean:

Risk level	What it means	Recommended action
 HIGH	Cases and risk are elevated — an outbreak may be developing.	Notify NCDC
 MEDIUM	Worth watching — some signs of rising activity.	Investigate
 LOW	Nothing unusual right now.	Monitor

4. **Send a warning (optional)** — click the “Notify NCDC” button to log an alert for every HIGH-risk state at once. Who receives it is explained in Section 4.5.

Demonstration diseases: If you pick a disease tagged “(demo)”, a clear yellow banner appears reminding you it is illustrative data, not real surveillance. See Section 5.

4.3 Register a Case — assess a patient in under a minute

This is the page you will use most. Work down the form:

1. **Add the symptoms** — in the “Symptoms” box, type to search and click each symptom that applies. Add as many as you can — the more you enter, the more reliable the assessment.
2. **Add anything missing** — if a symptom is not in the list, type it in the “Other symptoms” box, separated by commas.
3. **Fill in the patient details** — age group, sex, report date, and the state and LGA.
4. **Add extra detail if you have it (optional)** — open the “Environmental & exposure factors”, “Clinical notes”, “Laboratory information”, and “Supporting documents” sections to record things like flooding, rodent contact, lab/PCR results, or to upload a referral note.
5. **Click Predict** — the system reads the symptoms and shows an assessment.
6. **Read the assessment** — you will see the most likely condition, a confidence figure, a risk level, and a recommended action — plus an “Also consider” line with other possibilities.
7. **Save the case** — click “Save case”. The record is stored, and it also feeds the outbreak monitor, so the platform gets a little smarter with every case.

A wrong guess is worse than none: If the symptoms do not clearly point to one condition, the system will not force a diagnosis. Instead it says “Not a confident match yet” and lists the closest

possibilities, asking for more symptoms. You can still save the case for the record.

4.4 Trends — how things move over time

1. Choose what to look at — pick a Disease, a State, and what to show (Confirmed cases, Deaths, Outbreak probability, or Alerts).

2. Set the time span — drag the year slider to focus on a period.

Bars appear only for the months that actually have data — the platform never invents figures to fill empty months. If you choose a demonstration disease, the same yellow reminder banner appears.

4.5 Alerts — who was warned, and about what

Every warning the platform raises is logged here, newest first, with the disease, location, severity, message, who it went to, and how.

Warnings are addressed to a preset response list:

- **State Epidemiologist** — surveillance lead.
- **NCDC Duty Officer** — national coordination.

4.6 Model — what it covers and how it works

This page is the honest summary of the system's abilities. It shows the two models working together and lists exactly which diseases use real data and which use demonstration data. It is also the right page to show a reviewer who wants the detail — the plain-language version is here, and the technical version is in the Appendix.

5. Understanding what you see

Two jobs, two “layers”

It helps to know the platform works in two complementary layers:

- **Case triage** — the symptom checker on the Register page. It recognises 41 conditions from 132 possible symptoms, trained on a large international dataset (well beyond Nigeria), so it is broad.
- **Outbreak monitoring** — the state-by-state watch on the Outbreak Monitor and Trends pages. This needs real case-count history over time, which only exists for certain diseases.

Real data vs. demonstration data

So that the platform can show its full multi-disease ability while staying completely honest, the diseases it monitors fall into two clearly-marked groups:

Group	Diseases	What it means
Real data (4)	Lassa fever, Cholera, Mpox, COVID-19	Genuine surveillance records from official sources (NCDC/SORMAS,

Group	Diseases	What it means
		WHO/OWID and a peer-reviewed compilation). This is the credible core.
Demonstration (8)	Malaria, Typhoid, Measles, Yellow fever, Meningitis (CSM), Tuberculosis, Pneumonia, Diphtheria — each shown with a “(demo)” tag	Illustrative data used to demonstrate how the platform would monitor these diseases. It is clearly labelled everywhere and is NOT real surveillance data.

You will always know which is which: Demonstration diseases carry a “(demo)” tag in every list and chart, and selecting one shows a yellow banner. Nothing demonstration is ever presented as real.

Risk levels and confidence

Both pages use the same simple rule, so a label always matches its number:

- **HIGH** — 50% or more.
- **MEDIUM** — between 20% and 50%.
- **LOW** — under 20%.

On the Register page, “confidence” is how strongly the symptoms point to one condition. Below 50%, the platform holds back rather than risk a wrong answer, and asks you for more symptoms.

6. Everyday workflows

Short playbooks for the things you will do most:

A patient comes in feeling unwell

Go to  Register a Case → add their symptoms and details → Predict → read the assessment → Save.

“Is an outbreak starting anywhere?”

Go to  Outbreak Monitor → choose the disease → scan for  HIGH states → if needed, click Notify NCDC.

“Show me how this disease has been trending”

Go to  Trends → choose the disease and state → set the years → read the chart.

Let the platform watch on its own

“Who has been warned?”

Go to  Alerts → read the log → select any handled alert and click Mark acknowledged.

7. Frequently asked questions

Is the data real?

Yes for the four core diseases — Lassa fever, Cholera, Mpox and COVID-19 — which use genuine official surveillance records. Eight further diseases use clearly-labelled demonstration data so the platform can show its full range. You can always tell them apart by the “(demo)” tag.

Does it really e-mail officials?

It records every alert in the Alerts log by default. If it is connected to an e-mail server, it also sends the alert by e-mail and the method column will say “Email.” Without that connection, it keeps a complete in-app log.

Is it really “intelligent”? Can it run on its own?

Yes. It learns from every case you save (an adaptive model), and with Proactive alerting switched on it flags HIGH-risk states automatically, with little day-to-day input needed.

What diseases does it cover?

The symptom checker recognises 41 conditions. The outbreak monitor watches 12 diseases (4 on real data, 8 on demonstration data).

Does it work on a phone?

Yes — it runs in any web browser on a phone, tablet or computer. There is nothing to install.

Can it diagnose for certain?

No. It is a fast screening and early-warning aid. Always confirm with laboratory testing and clinical judgement before acting.

8. Good-practice tips

- Enter as many of the patient's symptoms as you can — it makes the assessment far more reliable.
- Always confirm a suggested diagnosis with laboratory testing before acting on it.
- Get the report date, state and LGA right — the outbreak monitor depends on them.
- Treat any “(demo)” disease as an illustration, not as a real situation on the ground.
- Acknowledge alerts once they have been handled, so the log stays meaningful.

Appendix — For the technical reviewer

This section is for readers who want the mechanics. Everyday users do not need it.

The models

- **Outbreak detection:** an adaptive online Support Vector Machine (SGD with a hinge loss and averaging, updated with `partial_fit`), with probabilities calibrated by Platt scaling. It is evaluated prequentially (test-then-train), reaching about 0.89 AUC — matching a train-once batch model while continuing to learn.
- **Case triage:** a Bernoulli Naïve Bayes classifier over 132 binary symptoms and 41 conditions, robust to partial symptom entry and able to update online. A confidence guard suppresses low-certainty predictions.

The data stores

- **Structured store (SQLite):** the case-count time series, registered cases, and the alert log.
- **Unstructured store (MongoDB):** clinical notes, laboratory information, and uploaded documents, linked to each case.
- **Live feed (Redis Streams):** new-case events for real-time processing. The two optional stores degrade gracefully when not configured.

Data sources (real diseases)

- Lassa fever — SORMAS / NCDC case-level line-list, per state.
- Cholera — a peer-reviewed sub-national Nigeria compilation (Charnley et al.).
- Mpox and COVID-19 — WHO / Our World in Data national series.